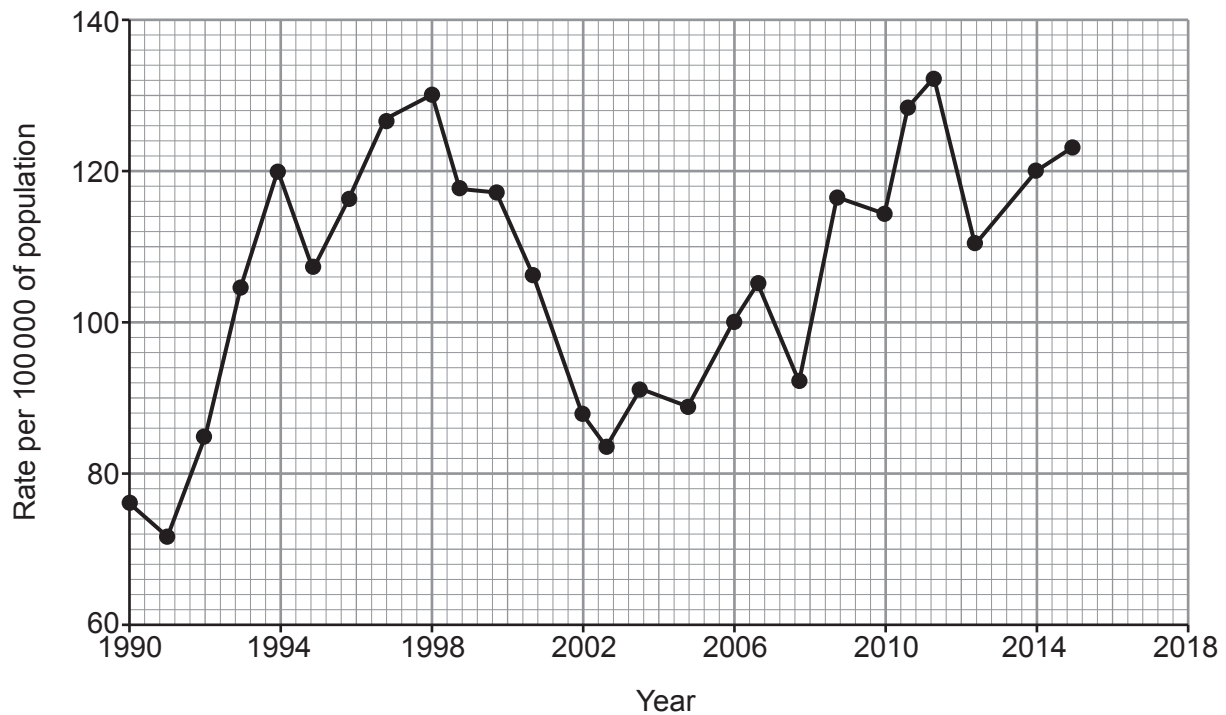


8. The graph below shows the confirmed cases of *Campylobacter* food poisoning per 100 000 of the population in Wales from 1990–2015.



In Wales, for every confirmed case of *Campylobacter* poisoning there are another two unconfirmed cases. The population of Wales is 3 million.

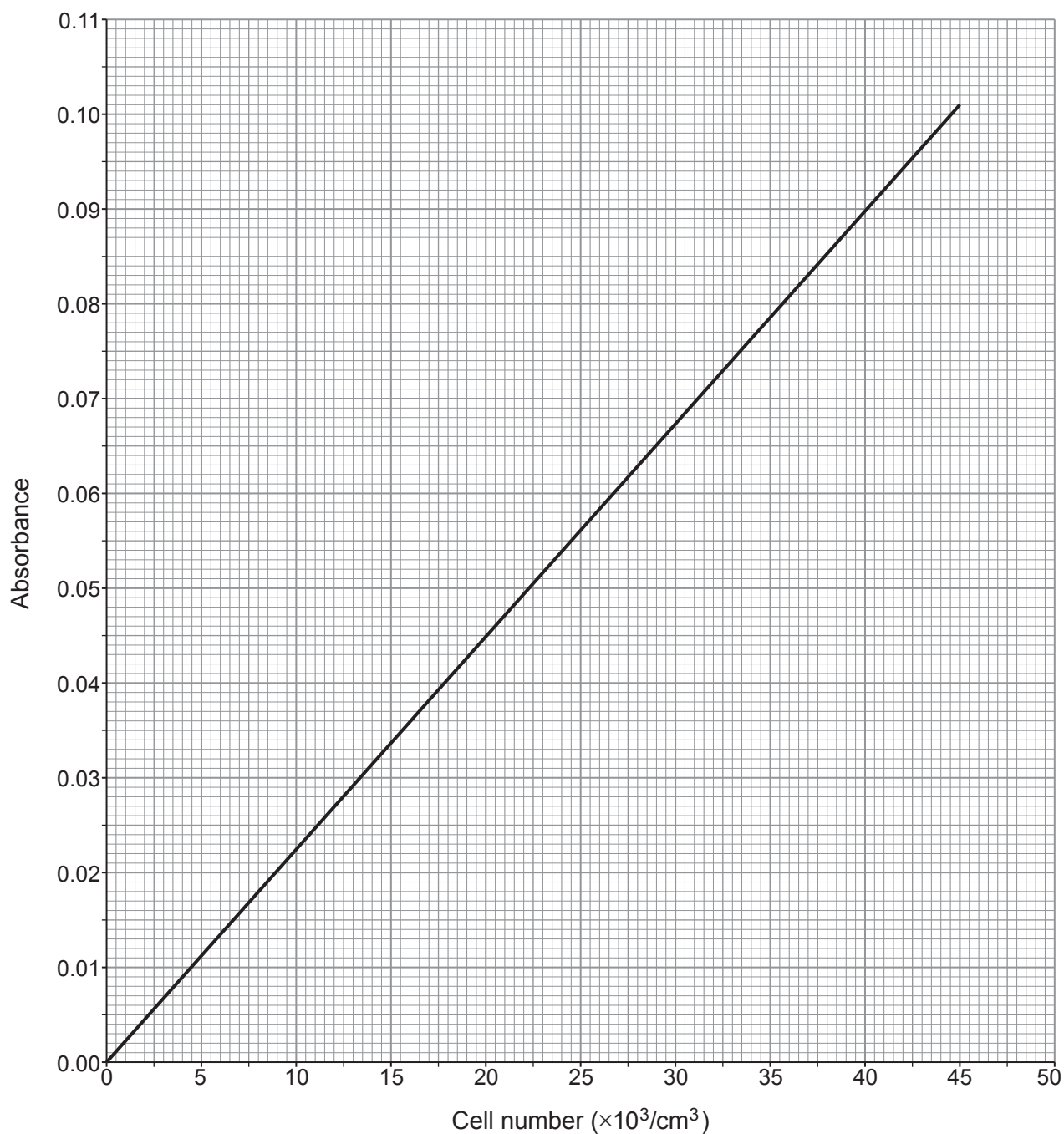
- (a) Calculate the total number of *Campylobacter* cases in Wales in 1998.

[3]

total number of cases =



- (b) Turbidity can be used to calculate the number of bacteria present in a sample. Alun used turbidity to study the growth rate of *Campylobacter* in water. He used the following standard graph.



(i) Alun analysed a water sample. The sample gave an absorbance of 0.01.

I. Use the graph to find the cell number per cm^3 at an absorbance of 0.01. [1]

cell number = per cm^3

II. *Campylobacter* reproduce every 20 mins The sample was left to stand for 1 hour. Alun thought that the absorbance would be 0.03 after 1 hour. Use your knowledge and data in the graph to explain whether he is correct. [3]

Space for calculations.

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(ii) The threshold value for making a person sick is 120 000 bacteria. In (b)(i)I. Alun's sample was 7 cm^3 . Use your answer above to determine whether this sample of water would make a person sick after standing for the hour. [3]

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END OF PAPER



Group

02

[illegible]

relative atomic mass

atomic number